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Accounting for the Korea Discount with Book-to-Market Ratios above One*

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ABSTRACT The historical average book-to-market ratio (BTM) of Korean listed companies is above one, and often used as evidence for the Korea discount. This paper asks whether the BTM reflects either risk inherent in Korean businesses as claimed in the popular press, or a bias in market expectations. Analyzing data of Korean listed firms from 1995 to 2019, we first confirm that stocks with BTM above one consistently outperform the rest, consistent with the empirical findings documented in prior studies. When we compose three summary indicators that are allegedly associated with the Korea discount (i.e., fundamental factor, geopolitical factor, and governance factor), we find that only governance factor is positively related to BTM as alluded to in the business press. In our subsequent analyses borrowed from Ohlson and Bilinski's (2015) specification, we find that BTM above one is associated with a higher probability of positive extreme returns, but a lower probability of negative extreme returns. The finding indicates that BTM above one is more likely to reflect characteristics associated with stock mispricing. We further explore why such systematic bias persists over time and report some evidence that governance factor plays a crucial role. This paper highlights the importance of recasting the underlying cause of a phenomenon.

KEYWORDS book-to-market ratio, Korea discount, risk explanation, mispricing explanation

I. Introduction

Korean stocks are traded at a discount (a.k.a., the Korea discount). The financial press reports that 1) the price-to-book ratios (an inverse of the book-to-market ratios) are on average below one in Korea and 2) are lower than those in other countries.¹⁾ Potential conduits include the fundamental risk inherent in an industrial structure (e.g., The Economist 2012; Yoon 2020), geopolitical risk (e.g., Kim and Park 2019; Jung et al. 2021; Pyo 2021), and governance risk (Forbes 2001, 2007; Rhee 2018; Ducret and Isakov 2020). It is unclear, however, whether and to what extent the book-to-market ratio (BTM hereafter) reflects the risk as claimed in the press (i.e., the risk explanation) because it could also manifest a bias in market expectation (i.e., the mispricing explanation). Given that the academic literature

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1) See articles in Business Korea (<http://www.businesskorea.co.kr/news/articleView.html?idxno=45899>), Maeil Business Newspaper (<https://www.mk.co.kr/news/english/view/2019/05/285314/>), and Korea Times (https://www.koreatimes.co.kr/www/tech/2021/07/693_47070.html).

does not converge on this issue (e.g., Ali et al. 2003), we assess the validity of relying on BTM in suggesting the widely accepted risk explanation for the Korea discount.

Analyzing data of Korean listed firms from 1995 to 2019, we first confirm that stocks with BTM above one consistently outperform the rest, consistent with the empirical findings documented in prior studies. Interestingly, when we compose three summary indicators that are allegedly associated with the Korea discount (i.e., fundamental factor, geopolitical factor, and governance factor), we find that only governance factor is positively related to BTM as alluded to in the business press.

More importantly, subsequent analyses based on Ohlson and Bilinski (2015) reveal that BTM above one is associated with a higher probability of positive extreme returns, but a lower probability of negative extreme returns. This finding contradicts the risk explanation that would predict higher probabilities for both positive and negative extreme returns, but instead supports the mispricing explanation.

We further explore why such systematic bias emerges and persists over time. We find that the mispricing captured in the Ohlson and Bilinski (2015) model is more salient in firms with higher governance factor. These results suggest that governance factor hinders market participants from correcting bias in a timely manner. In contrast, we find little evidence that either fundamental factor or geopolitical factor induces mispricing.

Our paper provides important contributions to the practice. First, we highlight the importance of recasting the underlying cause of a phenomenon. Practitioners have long assumed that BTM reflects the Korea discount. When we horserace two competing explanations (i.e., risk or mispricing), however, we find evidence supportive of the mispricing explanation that BTM reflects market failures of correctly assessing risk profiles of Korean companies. Second and more importantly, our findings have policy implications. Our results suggest that governance is indeed of a primary concern for the Korea discount. Our results would help set a priority of policy means to mitigate the Korea discount.

Our discussion also adds to the academic literature related to the return predictability of BTM. First, we extend the related literature by explicitly documenting the existence of the BTM effect in the Korean stock market. Second, we discuss the validity of BTM measures to describe Korean discount. Doing so, we belong to a rare stream of literature discussing on the Korean discount based on other measures such as price-to-earnings ratios and value-to-price ratios (Ducret and Isakov 2020; Cho et al. 2021).

We caution readers that our inferences are solely based on BTM. Although we do not

preclude that Korea discount itself would be a manifestation of market mispricing, our investigation provides little insight on this issue. Moreover, the business press often proposes low price-to-earnings ratios as alternative evidence of the Korea discount. Given that the price-to-book ratios additionally imply profitability (i.e., return-on-equity) above the price-to-earnings ratios (e.g., price-to-book ratio = return-on-equity $\times$ price-to-earnings ratio), we can infer the outcomes of each. In subsequent chapters, however, we strictly restrict ourselves to discussing the validity of using BTM for the Korea discount. Put differently, we test whether the BTM above one can be an effective heuristic.

II. Literature Review and Research Question

Numerous studies in finance document a positive association between the BTM and subsequent stock returns (e.g., Fama and French 1992; 1993), with two competing explanations. On the one hand, the BTM reflects a risk for which investors demand compensation. Fama and French (1992, 1993) argue that the return on the BTM compensates for the risk related to economic fundamentals, noting that high BTM stocks tend to have low earnings-to-assets ratios and low earnings persistence. Subsequent studies attempt to figure out the substance of risk reflected in the BTM, namely operating/financing risk (Penman et al. 2007), risk in earnings growth (Penman and Reggiani 2013), risk contained in the retained earnings (Ball et al. 2020), or macroeconomic risk (Petkova and Zhang 2005; Barth et al. 2021).

On the other hand, another stream of literature argues that the BTM-return association stems from a systematic mispricing of stocks with extreme BTM. They show that market participants underestimate future earnings for high BTM stocks and overestimate future earnings for low BTM stocks. Such systematically biased expectations are gradually corrected, leading to superior subsequent returns for high BTM stocks (La Porta et al. 1997; Skinner and Sloan 2002; Ali et al. 2003). Related studies suggest that arbitrage costs (Shleifer and Vishny 1997; Ali et al. 2003), investor sophistication (Lakonishok et al. 1994), or transaction costs (Knez and Ready 1996; Barber et al. 2001) explain why the systematic mispricing is not quickly and completely traded away.

Collectively, it is hard to conclude whether any noted factor related to the BTM is a risk, not only because the specified asset pricing model in Fama and French (1993, 1996) is based on an empirical analysis rather than on theory, but also because the observed BTM-return

relation may be due to mispricing (Penman et al. 2007). Yet, the discussion on the Korean discount pervasively assumes that the high BTM for Korean stocks is driven by risk profiles of Korean companies. This paper thus aims to document whether or not we can be confident in this assumption.

III. Data and Descriptive Statistics

3.1 Data and Sample

Our data includes Korean listed firms for the period from 1995 to 2019. From the initial sample, we exclude financial institutions and require that all variables are available for the analyses. We restrict the sample to firms with a December year-end. We also exclude firm-year observations with a negative book value. We obtain financial and stock market information from DataGuide. Ownership and governance information is obtained from the TS2000 database. The sources of other information used in analyses are indicated in the relevant part of the paper. The final sample comprises 30,712 firm-year observations.²⁾

3.2 Descriptive Statistics

Table 1 presents the descriptive statistics for the variables used in our analyses. The variable definitions are provided in the Appendix. The sample mean (median) of the book-to-market ratio (*BTM*) defined as the book value of equity divided by the market value of equity is 1.390 (1.063), confirming that Korean stocks are on average traded at a discount as featured in the financial press.

Figure 1 depicts time-series changes in *BTM* across the sample period. The left-hand side of the X-axis indicates the yearly *BTM* median while the right-hand side of the X-axis presents the yearly percentage of firms with *BTM* above one ($BTM > 1$) for all observations. We find that the medians of *BTM* generally exceed one before the mid-2000s and gradually decrease following the peak in 2008. The portion of $BTM > 1$ shows similar patterns. However, almost 50 percent of each yearly sample has a *BTM* above one, which is considerably higher than the percentage reported in the U.S. sample (Barth et al. 2021). Collectively, Figure 1 confirms the existence and persistence of the Korea discount phenomenon.

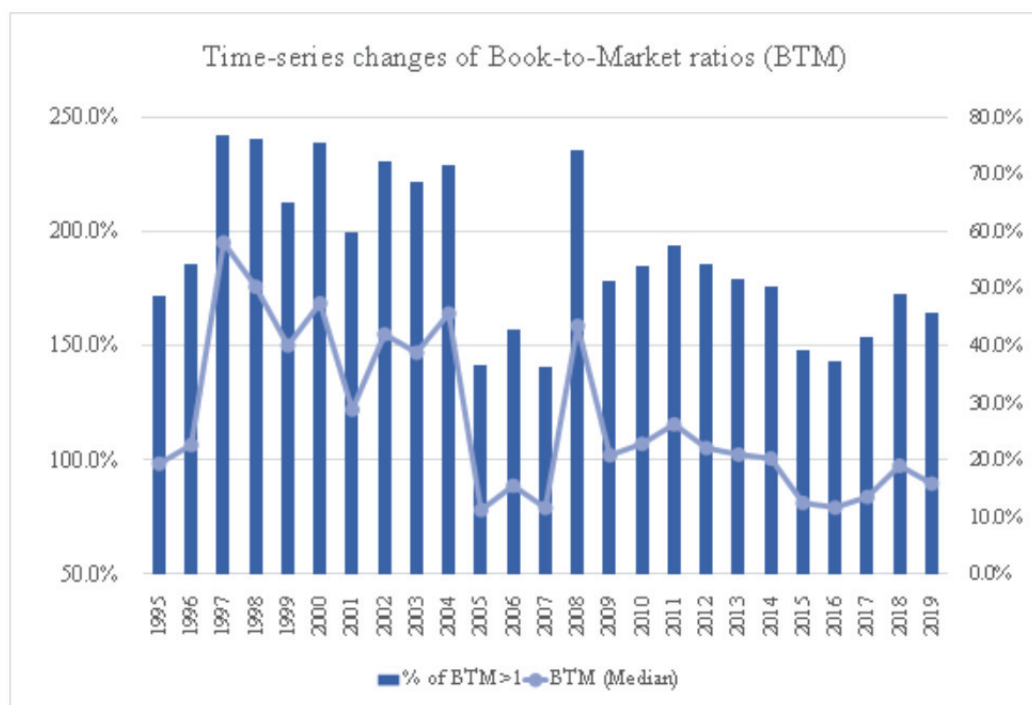
2) Due to the limited data availability, the sample size is reduced to 24,875 when government factor (*GOVFACTOR*) is included in the regression.

〈Table 1〉 Descriptive Statistics

Variable	N	MEAN	STD	P10	Q1	MED	Q3	P90
Return-related								
<i>CUMABRET</i>	30,712	0.058	0.671	-0.549	-0.273	-0.011	0.319	0.760
<i>HIGHABRET</i>	30,712	0.170	0.376	0.000	0.000	0.000	0.000	1.000
<i>LOWABRET</i>	30,712	0.116	0.321	0.000	0.000	0.000	0.000	1.000
BTM-related								
<i>BTM</i>	30,712	1.390	1.178	0.343	0.603	1.063	1.768	2.790
<i>LOG(BTM)</i>	30,712	0.779	0.409	0.295	0.472	0.724	1.018	1.332
<i>HBTM</i>	30,712	0.532	0.499	0.000	0.000	1.000	1.000	1.000
FACTOR-related								
<i>BETA</i>	30,712	0.991	0.700	0.184	0.562	0.963	1.392	1.837
<i>VOL</i>	30,712	0.180	0.104	0.086	0.113	0.153	0.215	0.300
<i>RECESSION</i>	30,712	0.064	0.244	0.000	0.000	0.000	0.000	0.000
<i>EPUNCERTAINTY</i>	30,712	0.000	1.000	-1.251	-0.862	-0.027	0.571	1.146
<i>FFACTOR</i>	30,712	0.000	1.000	-0.887	-0.565	-0.237	0.416	1.079
<i>GPFACTOR</i>	30,712	0.000	1.000	-1.021	-0.750	-0.171	0.268	2.160
<i>CHAEVOL</i>	24,875	0.119	0.324	0.000	0.000	0.000	0.000	1.000
<i>FOREIGNOWN</i>	24,875	0.061	0.104	0.000	0.002	0.015	0.066	0.192
<i>BODIND</i>	24,875	0.238	0.161	0.000	0.200	0.222	0.333	0.500
<i>GOVFACTOR</i>	24,875	0.000	1.000	-1.410	-0.249	0.295	0.515	1.091
Other Variables								
<i>LOG(MVE)</i>	30,712	18.128	1.521	16.369	17.102	17.946	18.908	20.112
<i>IFRS</i>	30,712	0.550	0.497	0.000	0.000	1.000	1.000	1.000
<i>CASH</i>	30,712	0.196	0.263	0.015	0.042	0.107	0.235	0.468
<i>RECEIVABLE</i>	30,712	0.571	0.785	0.052	0.127	0.307	0.677	1.339
<i>INVENTORY</i>	30,712	0.387	0.575	0.016	0.070	0.191	0.452	0.935
<i>PPENT</i>	30,712	1.126	1.661	0.066	0.214	0.551	1.289	2.707
<i>INTANGIBLES</i>	30,712	0.040	0.078	0.000	0.002	0.013	0.042	0.103
<i>GOODWILL</i>	30,712	0.018	0.065	0.000	0.000	0.000	0.000	0.038
<i>RD_CAPITALIZED</i>	30,712	0.007	0.023	0.000	0.000	0.000	0.000	0.019
<i>RD_EXPENSED</i>	30,712	0.024	0.043	0.000	0.000	0.006	0.029	0.072
<i>OTHERASSETS</i>	30,712	0.856	1.187	0.103	0.219	0.470	0.977	1.925
<i>LIABILITY</i>	30,712	1.878	3.074	0.148	0.343	0.856	2.012	4.346

Note: This table presents the descriptive statistics for the variables used in the analyses. Please refer to the Appendix for the variable definitions.

〈Figure 1〉 Distribution of the Book-to-Market ratios



Notes: This figure demonstrates the percentage of firms with *BTM* above one and the median of *BTM* by year.

IV. Empirical Findings

4.1 Determinants of BTM

We first explore whether the factors believed to be related to the Korea discount indeed determine *BTM*. Specifically, we estimate the following model:

$$\begin{aligned}
 BTM \text{ (HBTM)} = & \alpha_1 + \alpha_2 IFRS + \alpha_3 CASH + \alpha_4 RECEIVABLE \\
 & + \alpha_5 INVENTORY + \alpha_6 PPENT + \alpha_7 INTANGIBLES \\
 & + \alpha_8 GOODWILL + \alpha_9 RD_CAPITALIZED + \alpha_{10} RD_EXPENSED \\
 & + \alpha_{11} OTHERASSETS + \alpha_{12} LIABILITY \\
 & + \sum \alpha_{13 \sim 15} FACTORS + Year/Industry \text{ dummies} + \varepsilon .
 \end{aligned} \tag{1}$$

In Equation (1), we include the various determinants of BTM following prior literature (e.g., Barth et al. 2021; Cho et al. 2021). First, following Cho et al. (2021), we include a dummy variable that indicates the adoption of the International Financial Reporting Standard (*IFRS*) to control the effect of accounting standards. If the findings of Cho et al. (2021) that the gap between stock price and firm value has narrowed after the adoption of IFRS could be applied to book-to-market ratios as well, we expect a negative sign on α_2 . Next, we include several firm-level characteristics known to affect the book-to-market ratios (Barth et al. 2021). We control for cash and cash equivalents (*CASH*), net accounts receivables (*RECEIVABLE*), total inventory (*INVENTORY*), fixed assets (*PPENT*), total intangible assets (*INTANGIBLES*), the amount of goodwill (*GOODWILL*), total amount of R&D costs (*RD_CAPITALIZED* and *RD_EXPENSED*) and total liabilities (*LIABILITY*). To control the effect of other types of assets, we include *OTHERASSETS* which indicates assets other than those controlled above. All firm-level variables are scaled by the market value of equity at the beginning of the year (Barth et al. 2021).

To examine whether several factors discussed in the literature and media in relation to the Korean discount explain the Korea discount, we add three *FACTORS*: fundamental factor (*FFACTOR*), geopolitical factor (*GPFACOR*), and governance factor (*GOVFACTOR*) (The Economist 2012; Yoon 2020; Kim and Park 2019; Jung et al. 2021; Pyo 2021; Forbes 2001, 2007; Rhee 2018; Ducret and Isakov 2020). Specifically, *FFACTOR* is defined as the principal component of stock beta (*BETA*), return volatility (*VOL*), a dummy variable for market recession (*RECESSION*), and the economic policy uncertainty index (*EPUNCERTAINTY*) from Baker et al. (2016).³⁾⁴⁾ *GPFACOR* is an index of geopolitical situation in the inter-Korean relationship released by the website of Policy Uncertainty (https://policyuncertainty.com/korea_gpr.html).⁵⁾ *GOVFACTOR* is the principal component of several governance-related factors such as a dummy variable for business group (*CHAEVOL*), the percentage of foreign ownership (*FOREIGNOWN*), and board independence (*BODIND*).⁶⁾ *FOREIGNOWN* and *BODIND* are

3) Following French et al. (1987) and Fama and French (1993), we define stock market beta and return volatility as fundamental factors that affect stock prices.

4) The economic policy uncertainty index is obtained from the website of Policy Uncertainty (https://policyuncertainty.com/korea_monthly.html). We assign the average past 12-month monthly economic policy uncertainty index of each firm at the end of the fiscal year.

5) We use past 12-month monthly geopolitical situation index for each firm at the end of the fiscal year.

6) For *FFACTOR*, the eigenvalue of the first component is 1.387 and the first principal component of four attribute measures explains around 34.7% of the total variance. For *GOVFACTOR*, the eigenvalue of the first component is 1.621 and the first principal component of three different government factor proxies explains 54% of total variance.

multiplied by (-1) to indicate weak corporate governance. Detailed definitions of the variables are provided in the Appendix. We estimate Equation (1) using ordinary least square regression (logit regression) when the dependent variable is *BTM* (*HBTM*).

Table 2 shows the results of Equation (1). Columns (1) to (3) present the results with *BTM* as the dependent variable while columns (4) to (6) present the results with *HBTM*. These two sets of findings render similar inferences. The inclusion of *FACTORS* in the empirical model increases the adjusted (pseudo) R^2 , especially with the inclusion of *GOVFACTOR* (e.g., 0.836 in column (2) vs. 0.862 in column (3)), albeit with a smaller sample size due to data availability. More importantly, we find that *FFACTOR* and *GPFACOR* have negative associations with *BTM* and *HBTM*, inconsistent with public beliefs.⁷⁾ *GOVFACTOR*, however, has a positive relation with *BTM* as well as with *HBTM*. Thus, higher governance factor lowers the market valuation of equity, which in turn leads to higher *BTM*. Collectively, our findings suggest that governance factor can explain the Korea discount via *BTM*, whereas neither fundamental factor nor geopolitical factor does.⁸⁾

〈Table 2〉 Determinants of the Book-to-Market Ratio

Dep. Variables =	<i>BTM</i>	<i>BTM</i>	<i>BTM</i>	<i>HBTM</i>	<i>HBTM</i>	<i>HBTM</i>
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Intercept</i>	0.024 (0.80)	0.074* (1.83)	0.042 (1.22)	-9.445*** (-8.24)	-9.096*** (-8.80)	-9.058*** (-7.82)
<i>IFRS</i>	-0.129*** (-5.10)	-0.106*** (-4.39)	-0.080*** (-4.62)	-0.680*** (-5.12)	-0.566*** (-4.23)	-0.521*** (-3.53)
<i>CASH</i>	0.953*** (21.74)	0.951*** (21.83)	1.072*** (20.18)	9.078*** (9.50)	9.088*** (9.36)	9.510*** (8.65)
<i>RECEIVABLE</i>	0.585*** (10.36)	0.576*** (10.12)	0.594*** (7.03)	6.921*** (6.48)	6.832*** (6.45)	7.464*** (5.28)
<i>INVENTORY</i>	0.669*** (13.23)	0.662*** (13.02)	0.658*** (9.67)	7.421*** (6.91)	7.373*** (6.86)	7.457*** (5.63)
<i>PPENT</i>	0.635*** (18.38)	0.627*** (17.95)	0.661*** (12.79)	7.192*** (7.28)	7.135*** (7.21)	7.272*** (5.84)

7) Explaining *why* *FFACTOR* and *GPFACOR* are negatively loaded is beyond the scope of this study because we are interested in *whether* both risk indicators are positively associated with *BTM* as alluded to in the popular press. One possible explanation is that the significance of such risk may suppress book values to a greater extent relative to market values due to accounting conservatism (Barth et al. 2021).

8) When we divide the sample into two subgroups depending on *HBTM*, we find the magnitudes of coefficients on *FFACTOR*, *GPFACOR*, and *GOVFACTOR* are greater in the subsample with *HBTM*=1. These results imply that the *BTM*-return association would be greater in the high *BTM* group.

<i>INTANGIBLES</i>	0.880*** (7.54)	0.872*** (7.53)	0.676*** (5.84)	8.037*** (6.56)	7.977*** (6.64)	8.840*** (5.51)
<i>GOODWILL</i>	0.953*** (8.77)	0.953*** (8.88)	0.885*** (7.12)	9.089*** (6.56)	9.031*** (6.61)	9.900*** (5.59)
<i>RD_CAPITALIZED</i>	-0.164 (-0.58)	-0.141 (-0.50)	0.338 (1.31)	4.977*** (3.15)	4.980*** (3.16)	5.783*** (3.26)
<i>RD_EXPENSED</i>	0.124 (0.73)	0.066 (0.39)	0.026 (0.17)	-0.841 (-0.79)	-1.103 (-1.04)	-0.444 (-0.41)
<i>OTHERASSETS</i>	0.748*** (40.21)	0.739*** (38.61)	0.812*** (28.76)	8.338*** (8.80)	8.280*** (8.75)	8.754*** (7.46)
<i>LIABILITY</i>	-0.521*** (-12.70)	-0.513*** (-12.36)	-0.531*** (-7.92)	-6.451*** (-6.86)	-6.387*** (-6.79)	-6.713*** (-5.37)
<i>FFACTOR</i>		-0.166*** (-6.68)	-0.158*** (-4.70)		-0.991*** (-6.69)	-0.962*** (-5.54)
<i>GPFACTOR</i>		-0.072 (-0.94)	-0.852*** (-3.84)		-0.422 (-0.67)	-4.473*** (-2.95)
<i>GOVFACTOR</i>			0.022** (2.49)			0.110*** (2.68)
Fixed Effects	Industry, Year	Industry, Year	Industry, Year	Industry, Year	Industry, Year	Industry, Year
Observations	30,712	30,712	24,875	30,681	30,681	24,850
Adjusted (Pseudo) R ²	0.835	0.836	0.862	0.723	0.723	0.754

Notes: This table presents the results of Equation (1). Columns (1) to (3) (columns (4) to (6)) show the results with *BTM* (*HBTM*) as our dependent variable. We estimate Equation (1) using ordinary least square regression (logit regression) in columns (1) to (3) (columns (4) to (6)). *T*-statistics (*Z*-statistics) are presented in parentheses. *, **, *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively. Please refer to the Appendix for the variable definitions.

4.2 Does $BTM > 1$ outperform $BTM \leq 1$?

In this section, we examine whether *BTM* is related with subsequent stock returns as documented in the prior literature (e.g., Fama and French 1992, 1993). We specifically compare the future stock returns of $BTM > 1$ with those of $BTM \leq 1$. We measure future stock returns as the average monthly return over the next twelve months starting six months after the fiscal year-end (*ANNRET*). In column (1) of Panel A in Table 3, the average *ANNRET* for $BTM > 1$ is 0.021, meaning that firms with *BTM* above one report stock returns of 2.1% in the next twelve months on average. More importantly, such performance is superior to that for $BTM \leq 1$, which averages 0.007. The difference in mean values between these two groups is significant (t-stat=19.20, p-value<0.001) in general. However, the difference narrows over

time as presented in columns (2) to (6). The narrowing difference may suggest either that: 1) the risk premium on BTM decreases due to structural changes in risk profiles, or 2) the mispricing associated with BTM diminishes over time possibly due to arbitragers.

In Panel B, we further examine whether the return difference varies with *FFACTOR*, *GPFACOR*, and *GOVFACTOR*. Specifically, we divide the sample into two groups depending on whether each factor is above the sample median. We find that the return difference between stocks with $BTM > 1$ and those with $BTM \leq 1$ is significantly higher for firms with higher *FFACTOR* or higher *GOVFACTOR*. These findings suggest that $BTM > 1$ reflects both fundamental factor and governance factor and can be compensated for in subsequent stock returns. In stark contrast, however, the return difference is greater when *GPFACOR* is lower. This is inconsistent with the risk explanation. Collectively, the results in Table 3 suggest that both *FFACTOR* and *GOVFACTOR* can be potential conduits for the Korea discount.

〈Table 3〉 Future Returns of Firms with $BTM > 1$ and $BTM \leq 1$

Panel A. Full Sample

	Full Period	1995-1999	2000-2004	2005-2009	2010-2014	2015-2019
	(1)	(2)	(3)	(4)	(5)	(6)
Full Sample	0.015	0.031	0.015	0.007	0.014	0.015
$BTM > 1$	0.021	0.041	0.022	0.015	0.018	0.018
$BTM \leq 1$	0.007	0.012	-0.002	-0.001	0.009	0.012
Diff.	0.014***	0.029***	0.023***	0.016***	0.009***	0.006***
t-stat	19.20	6.93	11.70	10.78	7.52	4.76
p-value	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001

Panel B. Future returns depending on a cross-sectional variation in *FACTORS*

	<i>FFACTOR</i>			<i>GPFACOR</i>			<i>GOVFACTOR</i>		
	High	Low	High-Low	High	Low	High-Low	High	Low	High-Low
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
(1) $BTM > 1$	0.019	0.023	-0.004	0.020	0.022	-0.002	0.020	0.014	0.006
(2) $BTM \leq 1$	0.001	0.014	-0.013	0.011	0.002	0.009	0.007	0.006	0.001
Diff. (1-2)	0.018***	0.009***	0.009***	0.009***	0.020***	-0.011***	0.013***	0.008***	0.005***
p-value	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	0.001

Notes: This table compares the future stock returns between $BTM > 1$ and $BTM \leq 1$. Panel A presents the mean values of *ANNRET* for the full sample, firms with $BTM > 1$ and $BTM \leq 1$ for the full sample period and each sub-period. Panel B presents the mean values of *ANNRET* for each subgroup divided according to the level of *FFACTOR*, *GPFACOR*, and *GOVFACTOR*. *, **, *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively. Please refer to the Appendix for the variable definitions.

4.3 Risk or mispricing?

As aforementioned, two explanations exist for the BTM-return relations in prior studies. To distinguish which explanation is a better determinant of the Korea discount, we rely on the methodology developed in Ohlson and Bilinski (2015), which focuses on the signs of correlations between a trading attribute and extreme (i.e., high and low) returns. If the sign of the correlation coefficient between an attribute and a high return is opposite to that of a low return, it is more likely that the attribute-return relation is due to mispricing. Alternatively, if two correlations share the same sign, the risk explanation is more plausible. To follow this approach, we estimate the following two logit regressions:

$$P(HIGHABRET) = \alpha_1 + \alpha_2 VOL + \alpha_3 BETA + \alpha_4 LOG(MVE) + \alpha_5 LOG(BTM) + Year/Industry dummies + \varepsilon . \quad (2)$$

$$P(LOWABRET) = \alpha_1 + \alpha_2 VOL + \alpha_3 BETA + \alpha_4 LOG(MVE) + \alpha_5 LOG(BTM) + Year/Industry dummies + \varepsilon . \quad (3)$$

In Equations (2) and (3), *HIGHABRET* (*LOWABRET*) is defined as one if the twelve months cumulative market-adjusted return starting six months after the fiscal year end (*CUMABRET*) is higher (lower) than 50 (-50) percent. We include a set of risk controls, such as return volatility (*VOL*), market beta (*BETA*), and firm market capitalization (*LOG(MVE)*). The variable of interest is *LOG(BTM)*. *LOG(BTM)* is the natural log of *BTM*. All variables are defined in appendix.⁹⁾

Columns (1) and (2), Table 4 present the results of Equations (2) and (3), respectively. The coefficient on *LOG(BTM)* in column (1) is significantly positive (coefficient = 0.194, z-stat=3.97) while the corresponding coefficient in column (2) is significantly negative (coefficient = -1.199, z-stat=-15.60). These findings reveal that higher *BTM* is associated with a higher (lower) probability of positive (negative) extreme returns, which indicates that the BTM-return relation is supported by the mispricing explanation.

We further explore whether *FACTORS* known to be related to the Korea discount explain our findings. Specifically, we include and interact *FFACTOR*, *GPFACOR* and *GOVFACOR* with *LOG(BTM)* in Equations (2) and (3) and investigate which *FACTORS* derive the

9) Excluding *BETA* and *VOL* from the model does not alter our results.

observed results. The interaction terms are expected to have coefficients with the same signs as those of $LOG(BTM)$ in columns (1) and (2) if the interacted factor plays a role in the mispricing of BTM . In columns (3) and (4), we find evidence that the mispricing pattern observed in columns (1) and (2) are salient only when governance factor is higher. Collectively, our results in Table 4 indicate that the presence of $GOVFACTOR$ is mainly responsible for mispricing. To the extent that the weak governance increases the arbitrage risk resulting from the heightened transaction costs such as bid-ask spreads (e.g., Kanagaretnam et al. 2007), this interpretation is in line with the prior studies arguing that the BTM -return association reflects the mispricing in which the arbitrage risk plays an important role (Shleifer and Vishny 1997; Ali et al. 2003).^{10),11)}

〈Table 4〉 Ohlson and Bilinski (2015)'s Mispricing Test

Dep.Variable =	<i>HIGHABRET</i>	<i>LOWABRET</i>	<i>HIGHABRET</i>	<i>LOWABRET</i>
	(1)	(2)	(3)	(4)
<i>Intercept</i>	6.129*** (14.54)	-4.373*** (-7.73)	5.005*** (6.58)	-1.264 (-1.61)
<i>VOL</i>	-0.005 (-0.03)	3.324*** (14.82)	0.651 (0.92)	5.335*** (9.15)
<i>BETA</i>	-0.024 (-0.97)	0.083*** (2.82)	0.051* (1.87)	-0.031 (-0.94)
<i>LOG(MVE)</i>	-0.353*** (-18.01)	-0.054*** (-2.88)	-0.395*** (-16.43)	-0.105*** (-3.87)
<i>LOG(BTM)</i>	0.194*** (3.97)	-1.199*** (-15.60)	0.005 (0.07)	-1.307*** (-12.74)
<i>LOG(BTM)*FFACTOR</i>			0.007 (0.11)	0.017 (0.19)
<i>FFACTOR</i>			-0.392* (-1.73)	-0.278 (-1.56)
<i>LOG(BTM)*GPFACTOR</i>			-0.076 (-1.38)	-0.173 (-1.60)
<i>GPFACTOR</i>			-5.554*** (-4.36)	3.815*** (2.78)

10) We conduct intertemporal comparisons surrounding the Asian financial crisis in 1997 and the global financial crisis in 2008 confirm that our inference generally holds across the sample periods. Although the results are somewhat weaker in recent periods, BTM mispricing is salient in all sample periods.

11) When we compare KOSPI-listed firms and KOSDAQ-listed firms, we find that our results remain common in both samples but more salient for the former. This can be attributed to a higher average BTM of the former.

<i>LOG(BTM)*GOVFACTOR</i>			0.147**	-0.435***
			(2.09)	(-5.49)
<i>GOVFACTOR</i>			-0.109*	0.250***
			(-1.79)	(3.93)
Year FE	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
Observations	30,712	30,712	24,875	24,875
Pseudo R2	0.089	0.159	0.076	0.175

Notes: This table presents the results of Equations (2) and (3). The columns with odd (even) number contain the results for *HIGHABRET* (*LOWABRET*) as our dependent variable. In columns (3) and (4), we include the interaction terms between *LOG(BTM)* and each *FACTOR* (i.e., *FFACTOR*, *GPFACOR*, and *GOVFACTOR*) in Equations (2) and (3). Z-statistics are presented in the parentheses. *, **, *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively. Please refer to the Appendix for the variable definitions.

V. Conclusion

The financial press continues to use $BTM > 1$ as supporting evidence for the Korea discount. We test the validity of doing so. The novel approach of this paper is that we first construct three summary indicators for the factors allegedly reflective of the profiles of Korean businesses, namely fundamental factor, geopolitical factor, and governance factor. We then link these variables to $BTM > 1$ and find that only the governance factor is positively associated with $BTM > 1$. Next, we ask whether the positive return predictability of $BTM > 1$ is due to risk or mispricing. Our analyses based on Ohlson and Bilinski (2015) confirm that the return predictability of $BTM > 1$ is more likely due to mispricing than to risk. We also find evidence that governance factor is mainly responsible for this result. We conclude that governance factor primarily hinders market participants from correcting biases reflected in the BTM.

We hasten to point out that this study is by no means the ultimate answer to whether the Korea discount is due to risk or mispricing. Rather, with caution, our findings imply that the Korea discount is at least partly explained by the mispricing associated with the BTM. Throughout the analyses, we find that governance factor consistently plays a nontrivial role in explaining both the BTM and the return predictability of the BTM. By extension, our findings suggest that regulators and policy makers may be well-placed to leverage corporate

governance tools to resolve the Korean discount.

“This manuscript has not been published previously and is not under consideration by another publisher or journal, and the study in this manuscript has been conducted in accordance with the Ethical Guidelines set forth by the Korean Accounting Association.”

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〈Appendix〉 Variable Definition

Variable	Definition
Return-related	
<i>ANNRET</i>	= Average monthly stock returns over the 12 months starting from six months after the fiscal year-end.
<i>CUMABRET</i>	= The 12-month cumulative market-adjusted return starts six months after the fiscal year-end.
<i>HIGHABRET</i>	= An indicator variable that equals one if the <i>CUMABRET</i> is higher than 50 %, and zero otherwise.
<i>LOWABRET</i>	= An indicator variable that equals one if the <i>CUMABRET</i> is lower than -50 %, and zero otherwise.
BTM-related	
<i>BTM</i>	= The book-to-market ratio, defined as the book value of equity divided by the market value of equity.
<i>LOG(BTM)</i>	= Natural logarithm of the book-to-market ratio.
<i>HBTM</i>	= An indicator variable that equals one if the BTM is higher than one, and zero otherwise.
FACTOR-related	
<i>BETA</i>	= The market beta based on the past 36-month returns.
<i>VOL</i>	= A stock return's volatility, measured as the standard deviation of the monthly returns for the past 36 months.
<i>RECESSION</i>	= An indicator variable that equals one for fiscal years 1997, 1998, or 2008.
<i>EPUNCERTAINTY</i>	= Average of past 12-month monthly economic policy uncertainty index for each firm at the end of the fiscal year. The economic policy uncertainty score from https://policyuncertainty.com (Baker, Bloom, and Davis 2016)
<i>FFACTOR</i>	= Principal component of <i>BETA</i> , <i>VOL</i> , <i>RECESSION</i> , and <i>EPUNCERTAINTY</i> .
<i>GPFACTOR</i>	= Average of past 12-month monthly geopolitical situation index for each firm at the end of the fiscal year. Geopolitical risk score in the inter-Korean relationship from https://policyuncertainty.com (Baker, Bloom, and Davis 2016)
<i>CHAEVOL</i>	= An indicator variable that equals one for a business group (<i>chaebols</i>).
<i>FOREIGNOWN</i>	= Percentage of the ownership by foreign investors.
<i>BODIND</i>	= Board independence, measured as the percentage of independent directors.
<i>GOVFACTOR</i>	= The principal component of <i>CHAEVOL</i> , <i>FOREIGNOWN</i> , and <i>BODIND</i> . <i>FOREIGNOWN</i> and <i>BODIND</i> are multiplied by (-1) to indicate governance risk.
Other Variables	
<i>LOG(MVE)</i>	= Natural logarithm of the market capitalization.
<i>IFRS</i>	= An indicator variable for years after the IFRS adoption.

<i>CASH</i>	= Cash and cash equivalents in year t scaled by the lagged market value of equity.
<i>RECEIVABLE</i>	= Net accounts receivable in year t scaled by the lagged market value of equity.
<i>INVENTORY</i>	= Total inventory at the end of the year scaled by the lagged market value of equity.
<i>PPENT</i>	= Property, Plant, and equipment assets scaled by the lagged market value of equity.
<i>INTANGIBLES</i>	= Total intangible assets excluding goodwill and capitalized R&D scaled by the lagged market value of equity.
<i>GOODWILL</i>	= Goodwill scaled by the lagged market value of equity.
<i>RD_CAPITALIZED</i>	= Capitalized R&D scaled by the lagged market value of equity.
<i>RD_EXPENSED</i>	= Expensed R&D scaled by the lagged market value of equity.
<i>OTHEREASSETS</i>	Other assets, defined as total assets minus cash and cash equivalents, net = receivables, total inventory, property, plant, and equipment, intangible assets, goodwill, and capitalized R&D, scaled by the lagged market value of equity.
<i>LIABILITY</i>	= Total liabilities scaled by the lagged market value of equity.
